ABOUT ME

Samridhi Sharma — Personal, Academic, Career & Technical Profile

1. Personal Introduction

My name is Samridhi Sharma. I am a B.Tech Computer Science and Engineering student specializing in Artificial Intelligence and Machine Learning at Lovely Professional University (LPU), Punjab.

I am interested in building practical AI applications that solve real problems. My interests include Generative AI, Retrieval-Augmented Generation (RAG), Python development, AI automation, full-stack development, and hardware-integrated AI projects.

My journey is about more than learning technologies. I want to understand how systems work, build projects from scratch, explain them confidently in interviews, and eventually become a strong AI engineer.

I come from Kishtwar, Jammu and Kashmir, and I am working toward building my career in the technology industry, with Bengaluru as my preferred work location.

2. Education and Academic Background

- **University:** Lovely Professional University (LPU)
- **Degree:** Bachelor of Technology (B.Tech)
- **Branch:** Computer Science and Engineering
- **Specialization:** Artificial Intelligence and Machine Learning
- **Expected graduation:** 2027, according to my earlier academic details
- **School:** BVM Jammu
- **Class 10:** 89% in 2020
- **Class 12:** 79% in 2022

My academic journey has exposed me to programming, computer science fundamentals, artificial intelligence, and software development.

Academic detail to verify: Different academic records or conversations mention CGPA values of 4.72 and 6.29. I should confirm my latest official CGPA before using it in my resume or job applications.

3. My Career Vision

My main goal is to become a capable AI/ML or Applied AI engineer who can build useful, production-oriented applications rather than only theoretical models.

I am particularly interested in:

- Applied AI Engineer
- Generative AI Engineer
- AI Engineer Intern
- Python AI Developer
- LLM Engineer
- AI Automation Engineer
- Python Automation Engineer
- n8n Developer
- AI-focused software development roles

I want to work on real-world problems involving LLMs, document intelligence, intelligent automation, APIs, databases, and AI-powered products.

My longer-term compensation target is approximately **₹12 LPA**, while understanding that actual offers depend on my skills, interview performance, company, role, and market conditions.

Preferred location: Bengaluru, India.

I want to build a career in which my software engineering fundamentals and AI skills complement each other.

4. Professional Experience and Activities

Bluestock Fintech — Software Development Engineering Intern

January–February 2026 | Remote

My previously shared experience details include:

- Worked on more than 15 backend modules.
- Contributed to improvements associated with approximately 20% less redundancy.
- Worked with more than five REST APIs.
- Worked with a setup involving over 1,000 mock transactions.
- Reported 99.9% uptime for the relevant system or environment.
- Worked with more than 50 test cases and identified or addressed 15 bugs.

These are figures from my earlier resume-related discussions. I should verify their precise definitions and scope before presenting them as measurable professional achievements.

Divyanshi Sharma Podcast — Social Media Marketing

February 2025–present, according to previously shared details

- Worked on social media and audience growth.
- Previously reported more than 9,000 visitors over two months.

This experience demonstrates that my interests extend beyond coding into communication, content, audience engagement, and digital platforms.

Coding — Junior Internshala

January 2025, according to previously shared details

I have also mentioned a junior coding internship or experience through Internshala. The exact role title, responsibilities, and duration should be confirmed before inclusion in a formal resume.

Certification

- Automation Anywhere certification, dated 18 December 2025, according to my earlier information.

5. My AI/ML and Software Projects

I want my portfolio to demonstrate that I can move from an idea to implementation, testing, debugging, and explaining technical decisions.

Project 1: Ahaar AI — Packaged Food Health Scanner

Purpose: Help users understand packaged-food labels based on their dietary preferences and selected health considerations.

Technologies and features I have discussed:

- Python
- Kivy for a mobile-oriented interface
- MobileNetV2/CNN-based image classification
- Tesseract OCR for extracting text
- Food-label analysis
- User profiles for dietary considerations
- Alerts relating to sugar, sodium, and selected allergens
- A dataset previously described as containing 1,989 images across 13 classes

Previously discussed alert thresholds included sugar above 10 g and sodium above 150 mg. The applicability of these thresholds depends on serving size, product context, and the intended health rules.

I also reported a validation accuracy of 1.0 in an earlier experiment. That result needs careful validation for data leakage, class balance, and generalization before it can be presented as evidence of real-world model performance.

Personal motivation: I initially described this project as something I wanted to build for my mother. It connects my technical interests with a practical everyday use case.

Repository previously shared: `shokinw/Ahaar-AI`

Project 2: Autonomous RAG Research Assistant

Purpose: Allow a user to ask questions about research papers and receive answers grounded in retrieved document passages.

Technologies and components discussed:

- Python 3.10.11
- Ollama
- Llama 3.2
- LangChain

- FAISS
- Sentence Transformers
- PDF loading and text extraction
- Document chunking
- Embeddings and vector search
- FastAPI
- Streamlit

My earlier setup included a PDF-processing pipeline that produced 116 chunks.

The basic pipeline is:

1. Load a research PDF.
2. Extract its text.
3. Split the text into smaller chunks.
4. Convert chunks into embeddings.
5. Store embeddings in a vector index.
6. Embed the user's question.
7. Retrieve relevant chunks.
8. Give the retrieved context to the local language model.
9. Generate an answer based on the context.

I have been learning how retrieval quality depends on chunking, embeddings, query-document compatibility, and the relevance of retrieved passages.

A key goal is to make this system reliable, explainable, and usable without depending on paid model APIs.

Repository previously discussed: A repository under the GitHub username `shokinw`.

Project 3: AI Customer Support Escalator

An AI-oriented customer support concept intended to help identify support requests that may require escalation.

Relevant areas include text processing, classification, decision rules, customer-support workflows, and automation.

The exact implemented features and evaluation results should be documented separately.

Project 4: Smart Delivery Route Optimizer

Purpose: Explore route optimization for delivery operations.

Technologies previously discussed:

- React 19
- Vite
- Tailwind CSS
- Leaflet maps
- Node.js
- Express
- MongoDB Atlas
- Dijkstra's algorithm
- Render deployment

The central technical idea is to represent locations and connections as a graph and calculate an efficient path using a graph algorithm.

This project combines frontend development, backend APIs, database integration, maps, and algorithmic problem-solving.

Project 5: AI Hallucination Detector

An AI research or prototype project focused on identifying potentially unsupported or hallucinated model outputs.

I previously mentioned a result of approximately 92% over 1,000 or more samples. This should be treated as a preliminary, self-reported result until the dataset, evaluation methodology, baseline, and metrics are clearly documented.

The broader learning areas include factual consistency, verification, model evaluation, and the limitations of automated hallucination detection.

Project 6: AI Finance Tracker

Purpose: Build an AI-assisted personal finance tracking system.

Previously discussed components include:

- Samsung S23 FE as the intended mobile device
- PostgreSQL
- A database named `finance_tracker`
- SMS-based transaction synchronization
- Expense categorization
- An “Ask AI” interface
- Potential integration with payment-related notifications

My preference is to explore a free or locally controlled implementation wherever practical.

A production version would need careful permission handling, secure storage, accurate transaction parsing, and privacy protection.

Project 7: Second Brain

A personal knowledge-management assistant intended to organize information and make it searchable.

Previously discussed technologies:

- FastAPI
- PostgreSQL
- ChromaDB
- Ollama

The intended direction is to build a system that can store information, retrieve relevant content, and support question answering over personal notes or documents.

Project 8: Physical AI Research Assistant

I want to connect AI software with physical hardware by building a small research assistant with a display.

My plan is to develop the software first, test its functionality, and then integrate the hardware.

Previously discussed ideas include:

- A small OLED display, approximately 0.9 inches
- An ESP32 development board

- A local AI or RAG backend
- A question-and-answer interface
- A compact physical device for interacting with research content

I want to understand the complete project lifecycle: problem definition, architecture, software implementation, hardware integration, testing, debugging, and interview explanation.

Project 9: Turing-Inspired AI Project

I expressed interest in creating a beginner-friendly project inspired by Alan Turing's 1950 paper, *Computing Machinery and Intelligence*, particularly the imitation game.

My goal is to start with a small, understandable prototype and gradually expand it into a more sophisticated project without requiring paid APIs.

6. Hardware and Embedded Systems

Although I have focused heavily on software, I also want to explore hardware because it helps me understand how intelligent systems interact with the physical world.

I have worked with an Electrobot ESP-32S Wi-Fi IoT starter kit.

My ESP32 setup and debugging experience has involved:

- Arduino IDE
- USB serial connections
- COM ports in Windows
- CP210x and CH340 USB-to-serial drivers
- ESP32 board detection
- Upload and connection errors
- Serial communication
- GPIO pins
- Button and LED circuits
- `INPUT_PULLUP`
- Basic embedded programming

I previously tested or explored a setup involving GPIO18 for a button and GPIO21 for an LED.

I encountered upload errors such as connection timeouts and “No serial data received.” Debugging these issues has helped me learn that hardware projects require attention to cables, ports, drivers, board configuration, boot mode, and upload settings.

My broader ambition is to combine AI, embedded systems, and useful physical interfaces.

7. Technical Skills and Areas of Interest

The following represents my learning and project interests, not a claim that I have mastered every technology.

Programming

- Python
- C++
- JavaScript
- Object-oriented programming
- Data structures and algorithms
- Problem-solving and debugging

Artificial Intelligence and Machine Learning

- Machine learning fundamentals
- Neural networks
- CNNs
- NLP
- Text classification
- OCR
- Hugging Face
- TensorFlow/Keras
- scikit-learn
- OpenCV

Generative AI and LLM Applications

- Large language models
- Retrieval-Augmented Generation
- Embeddings

- Vector databases
- Chunking strategies
- Semantic search
- Prompt design
- Retrieval evaluation
- Hallucination reduction
- Local inference with Ollama
- LangChain
- FAISS
- ChromaDB

Backend and Full-Stack Development

- FastAPI
- REST APIs
- Node.js
- Express
- React
- Vite
- Tailwind CSS
- PostgreSQL
- MongoDB Atlas
- Webhooks
- API integration
- Deployment and debugging

Automation

- n8n
- Python automation
- Webhook-based workflows
- Google Sheets API
- AI-powered workflows
- Automation Anywhere
- Blue Prism
- CrewAI

Tools and Development Environment

- VS Code
- Git and GitHub
- Arduino IDE
- Docker
- Local AI tools
- Command-line workflows
- Streamlit

My goal is to connect these technologies into complete, working applications rather than study them as isolated topics.

8. Placement and Interview Journey

I have been preparing for campus placements and external opportunities while developing my technical portfolio.

My placement preparation has included:

- Coding assessments
- C++ output questions
- Pseudocode
- Data structures and algorithms
- Aptitude
- Technical interviews
- Group discussions
- Personal interviews
- Resume improvement
- Job applications
- Interview communication

I have discussed preparing for companies and assessments including Accenture, Cognizant, and other campus recruitment opportunities.

Previously discussed coding topics include:

- Arrays and hashing
- Two Sum

- Linked lists
- Next Greater Element
- Climbing Stairs
- House Robber
- Koko Eating Bananas
- Subsequences
- Arithmetic subarrays
- Straight-line and geometry problems
- C++ output and pseudocode questions

I have also prepared networking fundamentals, including:

- OSI model
- TCP versus UDP
- HTTP versus HTTPS
- Common network ports
- Networking devices

My assessment preparation has included an Accenture online-test opportunity referenced in September 2026. That was a historical opportunity, so its current status should not be assumed.

My placement experience

I have openly discussed facing multiple rejections during campus recruitment. At one point, I described a long sequence of unsuccessful drive outcomes and expressed frustration about being rejected despite investing years in learning.

These experiences have affected my confidence, but they also highlight an important goal: turning preparation into measurable improvement.

I want to become better at identifying the actual reasons behind unsuccessful attempts, improving my weak areas, and applying strategically instead of repeatedly restarting my preparation.

My focus is on becoming employable through demonstrable skills, solid projects, technical understanding, and better interview performance.

9. My GenAI and RAG Learning Journey

One of my current learning priorities is to understand RAG deeply enough to explain, implement, and debug it independently.

I want to be able to answer questions such as:

- What happens when a user submits a question to a RAG system?
- Why do we split documents into chunks?
- How do embeddings represent text?
- What does a vector database store?
- How does similarity search work?
- Why might retrieval return irrelevant chunks?
- How can chunk size and overlap affect results?
- How can I evaluate retrieval quality?
- What is the difference between retrieval errors and generation errors?
- How can I reduce unsupported answers?
- How do I connect the pipeline to FastAPI or a user interface?
- How do I run the system locally?

When retrieval fails, I want to investigate systematically:

1. Inspect the original document and extracted text.
2. Check whether the chunking process preserved useful context.
3. Examine which chunks were retrieved.
4. Verify the embedding model and query-document compatibility.
5. Inspect similarity scores and retrieval settings.
6. Test different chunk sizes or retrieval strategies.
7. Check the context passed to the language model.
8. Evaluate the final answer against the source document.

I do not want to memorize only definitions. I want to understand why a system behaves as it does and how to fix it when it fails.

10. My English and Communication Goals

Hindi is my first language, and I studied primarily through Hindi until Class 12. English is another language I have been developing.

I can often understand a topic and express my ideas, but I sometimes struggle to organize my thoughts into professional English, especially during interviews or spontaneous conversations.

I have discussed concerns about being judged for grammar, vocabulary, or fluency.

My goal is not to eliminate my accent or stop speaking naturally. My goal is to communicate clearly, confidently, and professionally.

I prefer:

- Simple English rather than unnecessarily complicated vocabulary
- Interview-focused speaking practice
- Natural conversation
- Feedback on grammar and word choice
- Better ways to express the ideas I already have
- Practice explaining technical projects
- Short, reusable answers for common interview questions
- Support in converting thoughts from Hindi into clear English

I want to improve my ability to explain my projects, describe technical decisions, participate in group discussions, and answer interview questions without freezing or overthinking.

11. My Learning Style

I learn best when explanations are practical, interactive, and broken into manageable steps.

My preferences include:

- Hinglish when a concept is difficult
- Simple English for technical explanations and interview practice
- Real-world examples
- Step-by-step instructions
- Complete code when implementation is required
- Dry runs for coding problems
- Clear explanations of compiler errors
- Comparisons between similar concepts

- Building small working examples before expanding them
- Debugging the actual problem rather than restarting everything

For coding practice, I prefer a process such as:

Try → Get stuck → Think → Write → Check → Correct → Understand.

I can become frustrated when preparation repeatedly returns to basic topics I already understand or when I have to restart from the beginning after covering several concepts.

I prefer continuity: remember what has already been covered, identify the exact gap, and move forward from there.

I also prefer honest feedback over exaggerated encouragement. I want to know what is working, what is weak, and what I should do next.

12. My Preferred Project-Building Approach

For a new project, I want a complete roadmap from the first step to the finished result.

My preferred sequence is:

1. Define the problem.
2. Decide what the first working version should do.
3. Identify the required tools and technologies.
4. Set up the environment.
5. Build a minimum viable version.
6. Test each component.
7. Integrate the components.
8. Debug actual failures.
9. Improve reliability and usability.
10. Document the architecture.
11. Prepare a GitHub repository and README.
12. Practise explaining the project in interviews.

I generally prefer free, local, and open-source tools, particularly when paid APIs are not necessary.

I use VS Code and terminal-based workflows frequently and appreciate exact commands and clear file locations.

13. Automation and Personal AI Systems

I am interested in using AI not just for demonstrations but also to automate everyday tasks.

One direction I have explored is an expense tracker using n8n, Docker, PostgreSQL, webhooks, and messaging integrations.

The broader idea includes processing transaction notifications, categorizing expenses, storing financial records, and asking questions about spending.

I have also explored ideas involving Evolution API, Gemini, WhatsApp-style workflows, and payment-related notifications.

My preferred design principles are:

- Minimize unnecessary recurring costs.
- Prefer local processing when practical.
- Keep workflows understandable.
- Use APIs and webhooks where appropriate.
- Store data in a structured database.
- Make automation easy to test and debug.
- Protect sensitive information.
- Avoid assuming that every integration is free or available without configuration.

14. Personal Interests and Everyday Goals

Apart from career development, I have discussed interests in:

- Technology news and AI developments
- News related to Jammu and Kashmir, India, and the world
- Short daily news routines
- Social media captions
- Instagram content
- Portfolio design
- Creative technology projects
- Practical AI tools
- Personal productivity

For social media captions, I tend to prefer short, natural, modern wording in English or Hinglish rather than formal or overly dramatic language.

For portfolio aesthetics, I have expressed interest in pink-and-blue themes, Poppins typography, Baloo 2, and visually clear statistics.

I also appreciate tools and routines that are free, easy to maintain, and realistic alongside college and placement preparation.

15. Personal Wellbeing Context I Have Shared

The following section records personal context shared in earlier conversations. These details are private and are not necessary for a public resume or professional portfolio.

I have discussed:

- Vegetarian food preferences and hostel limitations, including reliance on a kettle for some meals.
- A desire to gain weight through manageable nutrition and yoga without necessarily joining a gym.
- Skin that I described as dry, sensitive, combination, and acne-prone.
- Concerns about acne and irritation from some skincare products.
- A preference for gentler, affordable skincare options in India.
- Tired eyes and under-eye darkness that I associated with extensive laptop use and insufficient sleep.

These are personal observations and goals, not a medical diagnosis. Any health or skincare decisions should be based on appropriate professional advice when needed.

16. Strengths I Want to Build On

Based on the subjects I return to and the projects I want to build, several promising strengths stand out.

Curiosity: I ask how systems work, not only what they are called.

Practical orientation: I prefer working applications, integrations, and real examples.

Persistence: I continue exploring opportunities and technical projects even after setbacks.

Willingness to debug: I am prepared to investigate errors in code, local models, APIs, and hardware setups.

Interest in integration: I want to connect AI models with databases, APIs, interfaces, automation, and physical devices.

Self-awareness: I recognize areas where I need improvement, including interview communication, coding consistency, and technical depth.

These are qualities to develop further through consistent work, documented results, and feedback.

17. My Main Challenges

The challenges I have discussed most often include:

- Turning knowledge into strong assessment and interview performance.
- Maintaining coding practice without repeatedly revisiting the same basics.
- Improving aptitude, particularly profit-and-loss and cost-price/selling-price problems.
- Explaining technical concepts clearly under interview pressure.
- Building confidence in spoken English.
- Understanding debugging errors instead of applying fixes blindly.
- Balancing placements, projects, and deeper AI learning.
- Demonstrating project quality with reliable evaluation and clear documentation.
- Staying consistent after disappointing recruitment outcomes.

My objective is to approach these challenges as specific problems that can be worked on rather than treating them as proof that I cannot succeed.

18. My Current Priorities

My priorities, based on our earlier discussions, can be summarized as follows.

Career

- Focus on relevant Bengaluru opportunities.
- Apply for suitable AI, Applied AI, GenAI, Python, and automation roles.
- Tailor my resume to each role.

- Build a strong portfolio with demonstrable results.
- Prepare for technical interviews and coding assessments.
- Practise project explanations and professional communication.

Technical Learning

- Deepen my understanding of RAG and LLM application architecture.
- Strengthen Python, C++, DSA, APIs, databases, and debugging.
- Finish and document useful projects.
- Develop stronger evaluation methods for AI projects.
- Continue exploring local AI and hardware integration.

Communication

- Practise explaining my work in simple, clear English.
- Improve spontaneous interview responses.
- Develop concise introductions and project explanations.
- Practise technical and behavioural questions aloud.

Personal Development

- Maintain realistic study routines.
- Improve consistency rather than relying only on last-minute preparation.
- Balance career goals with rest and wellbeing.
- Review progress regularly and adjust the plan based on results.

19. Public Profiles and Portfolio References

The following handles were previously shared:

- **LinkedIn:** `samridhi-sharma-x07`
- **GitHub:** `shokinw`
- **LeetCode:** `samridhileet`

These handles should be checked for accuracy before they are published in a final portfolio or resume.

20. What I Want My Professional Identity to Represent

I want to become an engineer who can take an idea, understand the underlying problem, choose suitable tools, build the system, test it, debug it, and explain why it works.

I am particularly drawn to the space where AI meets software engineering and automation.

I do not want my professional identity to depend only on certificates, technology names, or claims about what I know. I want it to be supported by working projects, clear technical explanations, reliable results, and the ability to solve unfamiliar problems.

My journey is still in progress. I am learning, experimenting, correcting mistakes, and developing the skills required to move from student projects toward professional engineering work.

21. Accuracy and Privacy Notes

This document is based on information shared across previous conversations, not on an independent verification of every detail.

Before using it professionally:

- Confirm my latest CGPA and expected graduation date.
- Verify internship titles, dates, responsibilities, and measurable achievements.
- Validate all project performance figures and explain how they were measured.
- Confirm which projects are complete, deployed, or still prototypes.
- Update placement and application statuses because they change over time.
- Verify all public profile handles and repository links.
- Keep health, skincare, personal wellbeing, and other private details out of public versions unless I deliberately choose to share them.

Final summary: I am a CSE student specializing in AI and ML, building toward a career in Applied AI, Generative AI, and automation. I value practical learning, free and local tools, step-by-step technical guidance, working projects, and honest feedback. My goal is to turn my learning into strong engineering skills, a credible portfolio, better communication, and a successful career.